

Features: Useful Clues

Project: Helpful Classroom Sorter

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Categories and Labels

Without opening the old lesson, explain one key idea from Categories and Labels. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Pictures Become Numbers

Without opening the old lesson, explain one key idea from Pictures Become Numbers. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Input, Process, Output

Without opening the old lesson, explain one key idea from Input, Process, Output. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Prove that one measurement is insufficient for Features and useful distinctions. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Features and useful distinctions. Proposed explanation: “A stronger model can always recover an omitted measurement”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Design a record for a new task

A fictional display tray accepts a tile if its height is at most 9 cm and its thickness is at most 2 cm, including equality. Other dimensions are fixed. Before checking, you can record height, thickness and pattern name. A pass sticker is added afterward. Choose inputs with units. Construct two tiles with the same height but different labels. Then label a 9 cm high, 2 cm thick striped tile and explain why the sticker is unsuitable as an input before checking.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

A dataset contains only small smooth leaves and large toothed leaves. A model separates them by size. Design two fresh examples that distinguish a size shortcut from an edge-based rule.

6. Solve the given case

Two green leaves have smooth and toothed edges. The task is to distinguish edge types.

Choose the relevant feature and explain why size is insufficient.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

A sorter separates round and square tiles. Both shapes come in red and blue. Choose a useful feature and a distracting feature. Test your choice on a blue circle.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 8 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Select the smallest useful features for lost-and-found sorting and reject clues tied to a particular classroom or owner.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

A dataset contains only small smooth leaves and large toothed leaves. A model separates them by size. Design two fresh examples that distinguish a size shortcut from an edge-based rule.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Separate inputs from the target

For the fixed-orientation card task, fits requires length at most 12 cm and width at most 8 cm, including equality. Choose the measurements needed as inputs from length, width, colour, card number and the final inspection sticker. State the target label and write the feature record for a 12 cm by 8 cm red card using a defined order.

My reasoning and evidence

13. W2 Prove that one measurement is insufficient

Invent two cards with the same width but different correct labels under the card guide. Give both lengths and widths and justify the labels. Explain why a system receiving only width cannot always label both correctly.

My reasoning and evidence

14. W3 Challenge an apparent colour shortcut

A practice set contains three blue cards that fit and three red cards that are too large. A helper uses blue means fits and red means too large. Create a new card that this helper mislabels. Explain why the perfect practice score did not establish that colour was a reliable feature for this task.

My reasoning and evidence

15. W4 Design a record for a new task

A fictional display tray accepts a tile if its height is at most 9 cm and its thickness is at most 2 cm, including equality. Other dimensions are fixed. Before checking, you can record height, thickness and pattern name. A pass sticker is added afterward. Choose inputs with units. Construct two tiles with the same height but different labels. Then label a 9 cm high, 2 cm thick striped tile and explain why the sticker is unsuitable as an input before checking.

My reasoning and evidence

16. W5 Notice when the task changes

The delivery station now sorts cards into red and blue bins instead of checking dimensions. Which previously unneeded feature becomes useful? Explain why length and width alone may now be insufficient. Give two cards that support your explanation.

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Features: Useful Clues

A fictional shelf accepts a craft box only if width is at most 6 cm and height at most 4 cm.

Boxes P=(6,4,stripes), Q=(7,4,stripes), R=(6,5,dots), S=(6,4,dots).

A green acceptance sticker is attached only after the shelf check.

AI ACADEMY / AI-1/8

Dimensions determine shelf acceptance

	Width cm	Height cm	Outcome
P stripes	6	4	fits
Q stripes	7	4	width fails
R dots	6	5	height fails
S dots	6	4	fits

Read a cell using both its row and column labels.

Features: Useful Clues

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Width check

P,Q,R,S have widths 6,7,6,6.



2. Height check

Their heights are 4,4,5,4.



3. Apply both limits

P and S fit; Q fails width; R fails height.



4. Reject a shortcut

The later sticker cannot be an input available before checking.

CHANGE ONE THING

Change Q's width to 6 without changing its height.

Explain the changed result using the matching lesson.

Point to the pair that tests pattern, then explain why the sticker is a bad early input.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Choose sorter features that exist at decision time and actually connect to the stated classroom requirement.

19. Project portfolio milestone

Project: Helpful Classroom Sorter

Current milestone: Make a transparent first version. Keep a trace of how one input becomes an output; explain the rule or learned component.

Choose sorter features that exist at decision time and actually connect to the stated classroom requirement.

Evidence to keep · My record

Evidence to keep: This week’s input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week’s record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional shelf accepts a craft box only if width is at most 6 cm and height at most 4 cm.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? The later sticker cannot be an input available before checking.

1. Apply both limits
2. Reject a shortcut
3. Height check
4. Width check

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Reject a shortcut The later sticker cannot be an input available before checking. Comparing P with Q isolates width; comparing P with R isolates height. Comparing P with S changes pattern while both remain accepted. Under this declared policy, pattern is not a deciding feature. A later acceptance sticker leaks the very answer the system should produce.

Practice 2: explain the deciding evidence

Changed-case practice: Change Q's width to 6 without changing its height. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. This one example guarantees correct results for every future input.
3. Q fits the shelf.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Q fits the shelf. Both dimensions now meet their including equality limits; stripe pattern does not change the rule.

Repair a missing building block

Review Categories and Labels (ai-1/7). Goal: Apply a category definition using explicit evidence and boundary rules. Short practice: Explain why B originally was revise rather than review, then label its changed version.. Return to this week and explain what you changed.

Review Pictures Become Numbers (ai-1/5). Goal: Encode and reconstruct a small image using an explicit grid convention. Short practice: Find a position where the original badges differ, then read the changed A back as a picture.. Return to this week and explain what you changed.

Review Input, Process, Output (ai-1/2). Goal: Trace information through a system without confusing information with a physical action. Short practice: Use the richer record to find blue pencils in Box B. Explain why total alone was insufficient.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
